

ECE 3337: Test #2

Spring 2024

Last name of student	Student ID number
First name of student	Email address

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	

1. (20 points) Laplace Transforms

a. (6 points) Find $\mathbf{X(s)}$, the Laplace transform of the signal $x(t) = 2\delta(3t - 4) + tu(t - 1) + t^2e^{1-t}u(t)$.

(b) (6 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = 4e^{-t}x_2(2t - 2)$ in terms of $\mathbf{X_2(s)}$

c. (8 points) Find the Laplace transform of $z(t) = (t - 2) e^{-t} \cos(t - 2) u(t - 2)$, and find the values of all its poles and zeros.

2. (20 points) Inverse Laplace Transform

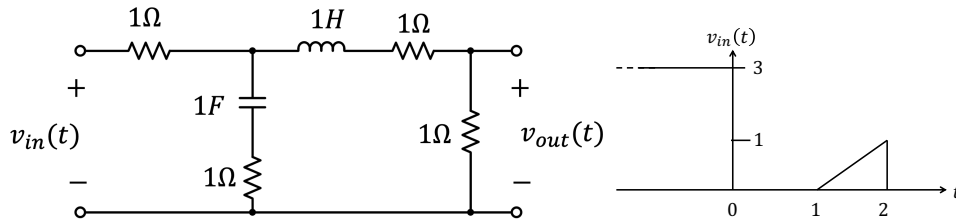
Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{e^{-3s}(2s+3)}{(s^2+9s+14)}$

b. (10 points) $X(s) = \frac{s^2+1}{(s+2)(s^2+3)}$

3. (25 points) Circuit Analysis

For the circuit below, the input $V_{in}(t) = x(t)$ is given in the graph. The units for $x(t)$ is Volts [V].



- (5 points) Draw the s -domain representation of the circuit with the initial conditions indicated.
- (5 points) Find the s -domain representation of the input $v_{in}(t)$ for $t > 0$.
- (10 points) Write the mesh current equations in the s -domain needed to find the output $V_{out}(s)$. Express your answer by filling in the entries in the Cramer's method matrices shown below. Here, $\mathbf{I}_1$ and $\mathbf{I}_2$ represent the mesh currents (left to right), and $\mathbf{V}_1$, $\mathbf{V}_2$ represent mesh voltages.

$$\begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{A}_3 & \mathbf{A}_4 \end{pmatrix} \begin{pmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{pmatrix}$$

d. (5 points) Write the formula for $V_{out}(s)$ in terms of $V_{in}(s)$.

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

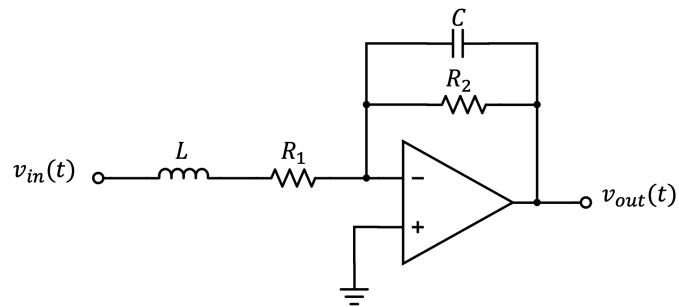
$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = x(t)$$

The input is $x(t) = e^{-t}u(t)$, with the initial conditions $y(0-) = 3$, $y'(0-) = 1$.

- a. (5 points) Derive a formula for the output $\mathbf{Y}(s)$. Your answer should be expressed as a ratio of polynomials with a fully factored denominator.
- b. (5 points) Find the modes of the system represented by the above LCCDE. Calculate the natural response of this system in the Laplace domain.

- c. (5 points) Calculate the initial value $y(0+)$ of this system's response.

5. (10 points) Operational Amplifiers



- a. (6 points) Calculate the transfer function $\mathbf{H(s)}$ for this circuit.
- b. (4 points) Identify the poles and zeros of the transfer function $\mathbf{H(s)}$.

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.

